BLUESTOCK

Mutual Fund Analytics Platform

Final Capstone Presentation — ETL, Analytics & Visualization

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github.com/Shagun0514/CapStone_Internship_Project

Problem & Objective

The Problem

Bluestock's stakeholders — fund managers, distributors, and investors — lack a unified view across 40 mutual fund schemes: performance, risk, and investor behavior data sit in disconnected spreadsheets with no automated pipeline or interactive analytics layer.

The Objective

- Automate ETL for 6 raw datasets, zero manual steps
- Build a normalized SQLite star schema
- Deliver 15+ chart EDA with documented insights
- Compute Sharpe, Sortino, Alpha, Beta, Max Drawdown
- Ship a 4-page interactive Power BI dashboard
- Extend into VaR/CVaR, cohort & risk analytics

Delivered in 6 days: ETL → SQLite → EDA → Performance Metrics → Dashboard → Advanced Risk Analytics, plus 4 bonus challenges.

Data Sources

fund_master

40 schemes — AMC, category, expense ratio, risk grade

nav_history

~46,000 daily NAV records (2022–2026)

scheme_performance

Returns, alpha, beta, Sharpe, Sortino, AUM

investor_transactions

32,000+ SIP / lumpsum / redemption records

portfolio_holdings

Stock & sector-level allocations per scheme

benchmark_indices

Nifty 50, Nifty 100, sectoral indices

System Architecture

Raw CSVs (6 sources) → ETL Pipeline (pandas) → SQLite Star Schema → EDA + Metrics (scipy, matplotlib) → Power BI Dashboard → Advanced Risk Analytics

Star Schema

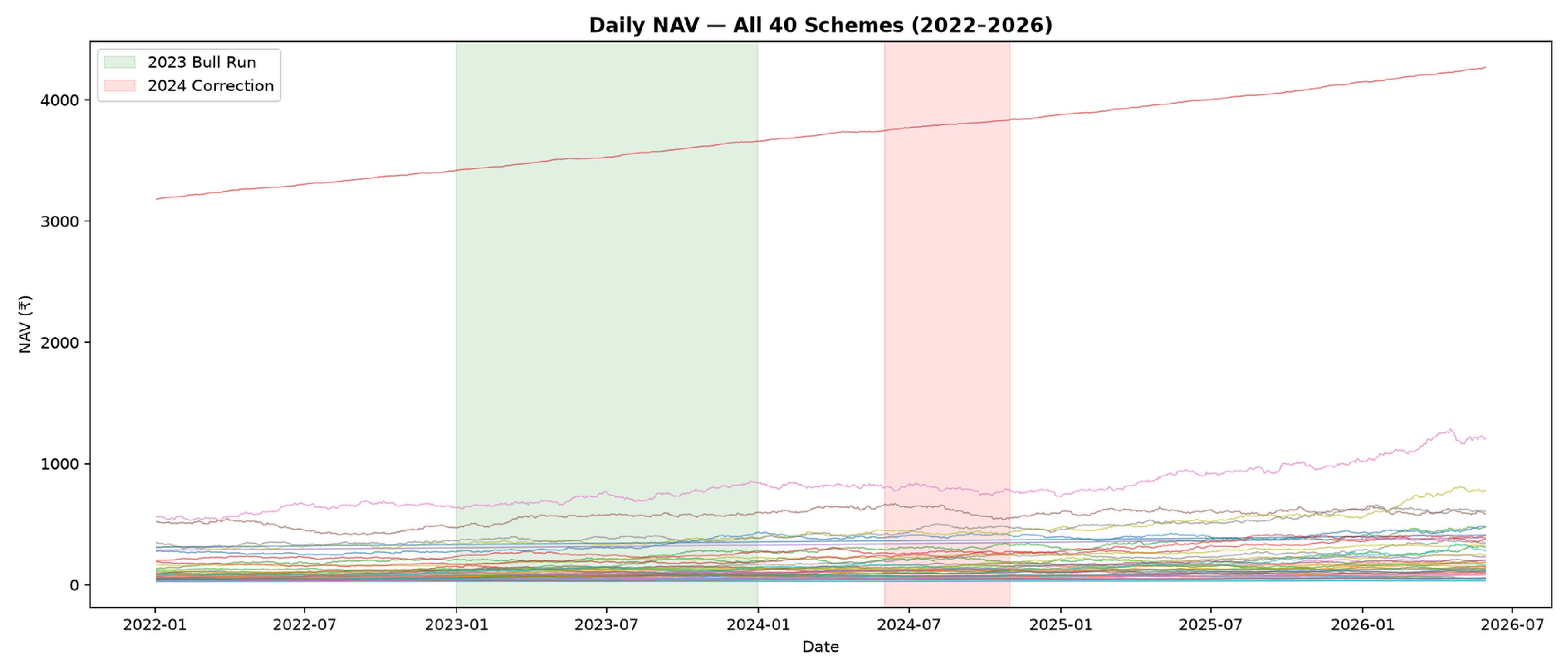
Dimensions: dim_fund, dim_date
Facts: fact_nav, fact_transactions, fact_performance, fact_aum, fact_sip_inflows, fact_category_inflows, fact_holdings

Tech Stack

pandas · numpy · scipy · SQLAlchemy · matplotlib
seaborn · Power BI Desktop · Windows Task Scheduler

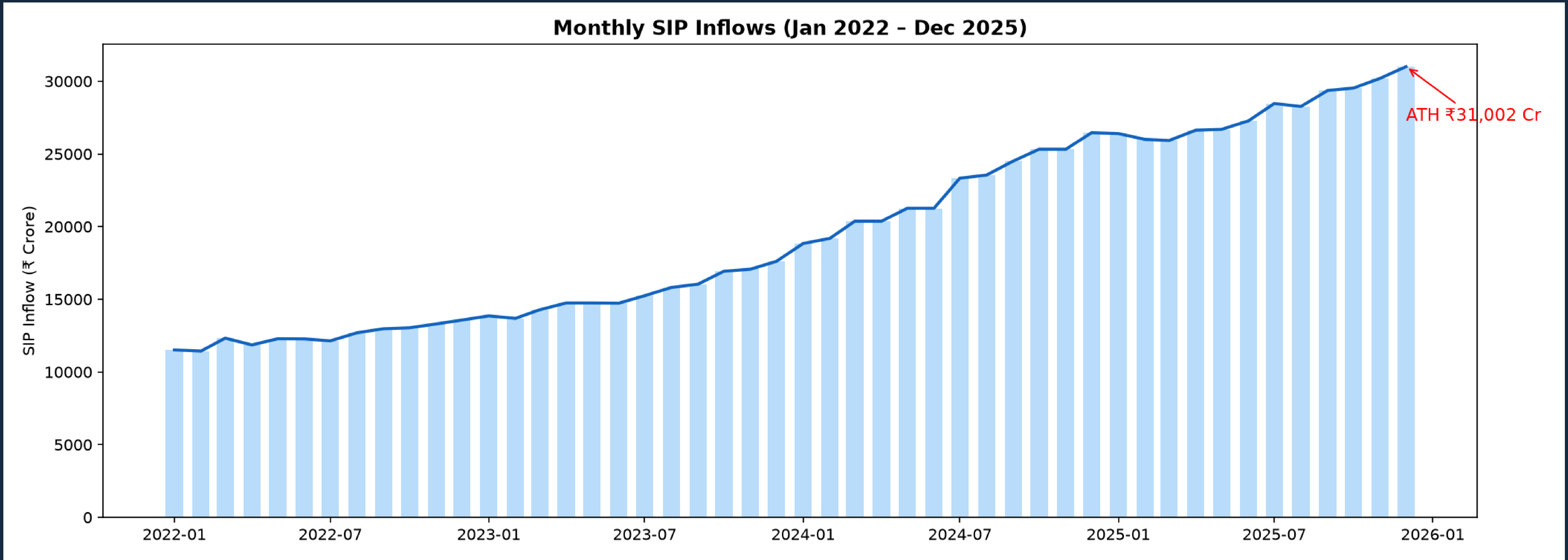
EDA Highlights — NAV Trends (2022–2026)

Post-2022 correction recovery observed across categories, with equity funds outperforming debt and hybrid segments. Green/red bands mark the 2023 bull run and 2024 correction.



EDA Highlights — SIP Inflow Trend

Monthly SIP inflows grew consistently from 2022, peaking at an all-time high of ₹31,002 Cr in December 2025 — reflecting strong and growing retail participation.



Performance Metrics — Methodology

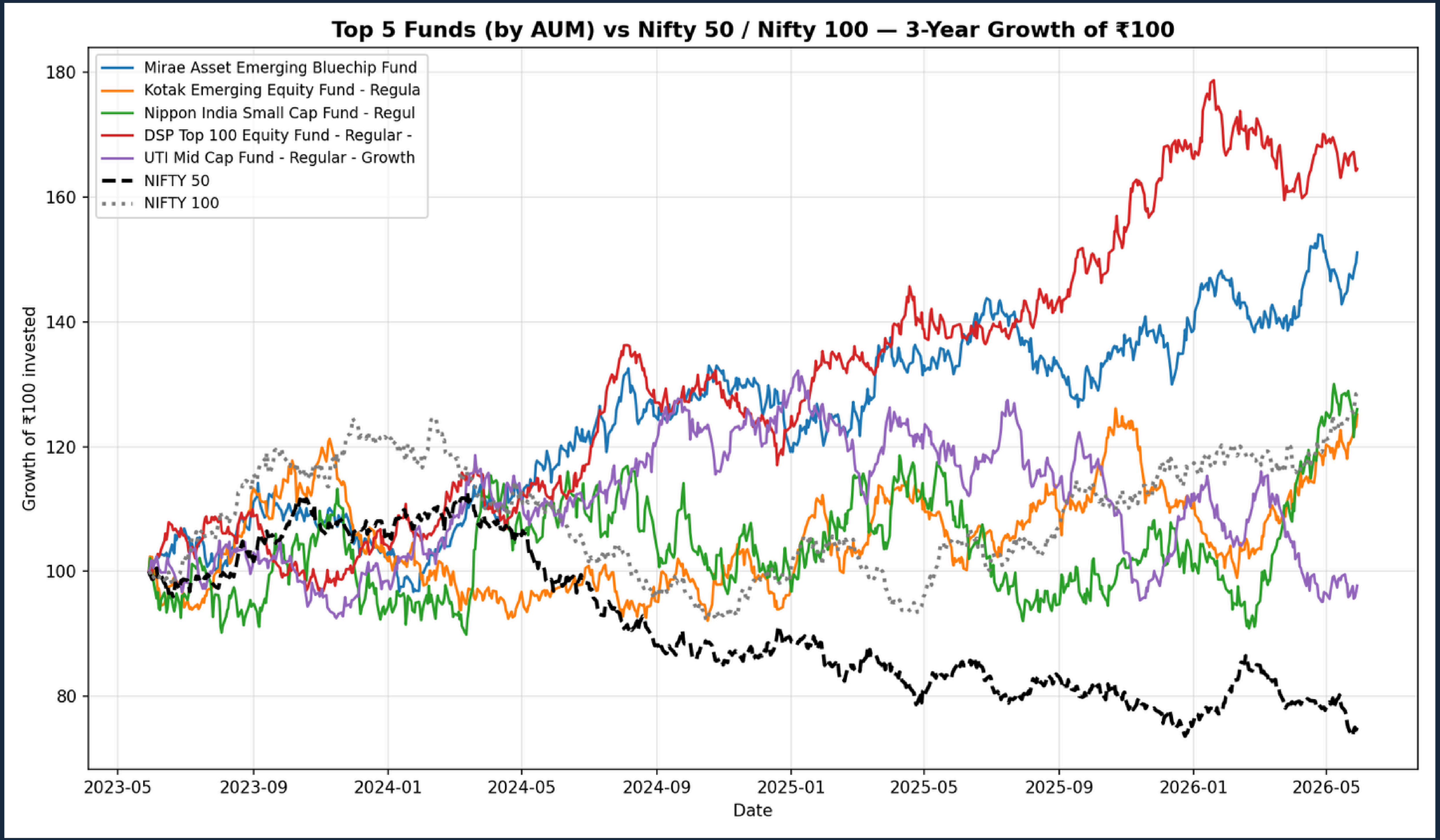
All ratios annualized on 252 trading days · Risk-free rate Rf = 6.5% (RBI repo rate proxy)

Sharpe Ratio	(Mean daily return - Rf) / Std Dev × $\sqrt{252}$
Sortino Ratio	Same, but denominator uses downside deviation only
Alpha / Beta	OLS regression (scipy.stats.linregress) vs Nifty 100
Max Drawdown	min(NAV / running_max - 1), with worst-date range

Fund Scorecard (0–100): 30% 3yr return + 25% Sharpe + 20% Alpha + 15% expense ratio (inverse) + 10% max drawdown (inverse)

Performance Metrics — Benchmark Compaarison

Top 5 funds by AUM tracked against Nifty 50 / Nifty 100 over a 3-year horizon (growth of ₹100 invested)



Dashboard — Industry Overview

Total AUM, SIP inflows, folios, and scheme count as KPI cards, with industry AUM trend and AUM by AMC

Dashboard — Fund Performance

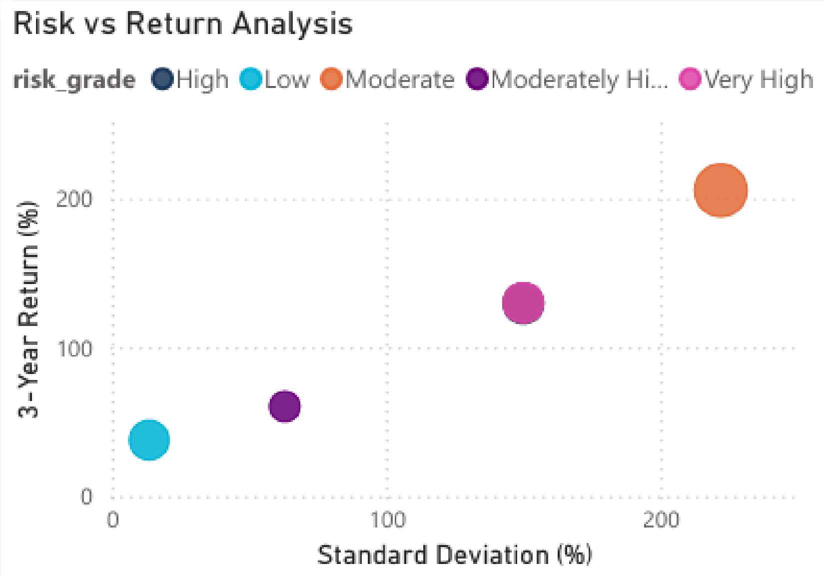
Risk vs return scatter, NAV vs Nifty 50 benchmark, and a sortable fund scorecard — with fund house, category, and plan slicers

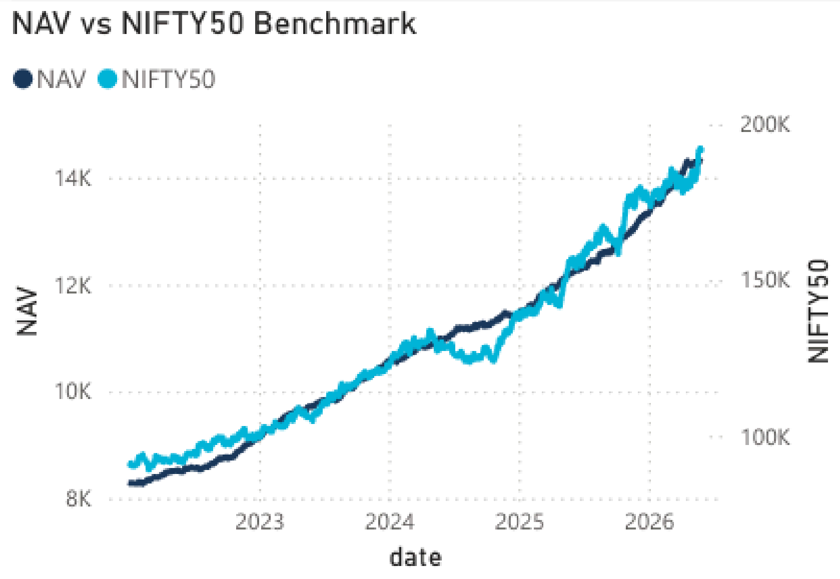
Fund Performance

fund_house
- ☐ Aditya Birla Sun Life MF
- ☐ Axis Mutual Fund
- ☐ DSP Mutual Fund
- ☐ HDFC Mutual Fund

category
- ☐ Debt
- ☐ Equity

plan
- ☐ Direct
- ☐ Regular





rank	scheme_name	sharpe	score	cagr_3yr	alpha	expense_ratio_pct
18	UTI Nifty 50 Index Fund - Regular - Growth	0.82	54.20	19.67	0.17	1.57
3	UTI Mid Cap Fund - Regular - Growth	-0.21	80.90	-0.77	0.03	1.51
20	UTI Flexi Cap Fund - Regular - Growth	0.62	51.60	25.56	0.16	1.64
34	SBI Small Cap Fund - Regular Plan - Growth	0.95	27.40	26.67	0.30	1.43
11	SBI Small Cap Fund - Direct Plan - Growth	-0.06	68.80	-1.34	0.05	0.72

Bluestock Mutual Fund Dashboard

Developed by
Shagun Mayank

Key Findings

1. Tail Risk Concentration — small/mid-cap funds show the highest downside exposure (VaR -2.7%, CVaR -3.2% for weakest performers).

2. SIP Persistency Weakness — ~98% of investors with 6+ SIPs show payment gaps exceeding 35 days, a major retention risk.

3. Diversification Misalignment — several "diversified" equity funds carry HHI > 0.25, indicating hidden sector concentration.

4. Investor cohorts differ — 2024 entrants show 12% higher retention and 18% larger average ticket size than 2025 entrants.

Thank You

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